

AI TRAVEL PLANNER

A course project report submitted in partial fulfilment of the requirement

of

DEVOPS AND FULLSTACK

By

P.LIKITHA -2303A52393

RUPALI -2303A52308

G.SRINIDHI -2303A52285

Under the guidance of

Dr. N.VENKATESH

Asst. Professor, Department of CS & AI



SR University, Warangal

APRIL-2026

CONTENTS

ABSTRACT	03
1 INTRODUCTION	
1.1 About the Project	04
1.2 Objectives	04
2 LITERATURE REVIEW	
2.1 Overview of DevOps and Full-Stack Development	05
2.2 Related Studies and Existing Tourism Applications.....	05
2.3 Tools and Technologies Used	05
3 METHODOLOGY	
3.1 System Architecture	06
3.2 Data Handling and Authentication.....	06
3.3 Functional Modules	06
4 RESULTS AND DISCUSSION	
4.1 Implementation Results	07
4.2 Key Features Achieved	13
4.3 Discussion	13
5 CONCLUSION AND FUTURE WORK	
5.1 Conclusion	14
5.2 Future Scope	14

ABSTRACT

This project presents the development of a full-stack web application named “AI Travel Planner”, designed to simplify and enhance travel planning by providing users with an intelligent and interactive platform. The application enables users to explore travel destinations, famous places, temples with historical significance, budget-friendly hotels, and top-rated restaurants in a structured and user-friendly manner.

The system incorporates user authentication (signup/login) to personalize the user experience. Users can input travel details such as destination, budget, and number of days, and the application generates a complete AI-based travel plan including day-wise itineraries without duplication. The frontend is built using modern web technologies to provide a responsive and visually appealing interface, while the backend is developed using Node.js, Express.js, and MongoDB for secure data management.

Additionally, the platform includes budget optimization, hotel booking, and email features, ensuring a seamless and efficient digital travel planning experience.

CHAPTER 1: INTRODUCTION

1.1 About the Project

“AI Travel Planner” is a full-stack web application designed to simplify and enhance travel planning using artificial intelligence. It provides users with an interactive platform to explore destinations, generate personalized travel plans, and manage their trips efficiently.

The application allows users to enter travel details such as destination, budget, and number of days. Based on this input, the system generates a structured travel plan including day-wise itinerary, famous places, temples with historical significance, budget-friendly hotels, and top-rated restaurants.

The application includes:

- User registration and login system
- Plan trip page (destination, budget, days)
- AI-generated travel plan
- Temples with history and location
- Famous places and attractions
- Hotel recommendations with booking feature
- Restaurant suggestions
- Email feature to send trip details

1.2 Objectives

- To develop a full-stack AI-based travel planning application
- To implement secure login and signup functionality
- To generate personalized travel plans using AI
- To optimize travel budget effectively
- To provide temple history and cultural insights
- To enable hotel booking within the application
- To create a responsive and user-friendly interface

CHAPTER 2: LITERATURE REVIEW

AI-Based Travel Planning and Full-Stack Development

Modern travel applications increasingly rely on artificial intelligence and full-stack development to deliver personalized and efficient user experiences. AI helps in analyzing user preferences and generating customized travel recommendations.

Frontend frameworks like React enable dynamic and responsive user interfaces, while backend technologies such as Node.js and Express.js support scalable API development. Databases like MongoDB provide flexible and secure data storage.

Authentication mechanisms such as JWT and bcrypt ensure secure user access. Additionally, integrating AI with travel systems enhances user experience by providing intelligent suggestions such as temples, historical places, hotels, and restaurants.

Studies show that combining AI with full-stack systems improves efficiency, reduces manual effort, and enhances user satisfaction in travel planning platforms.

CHAPTER 3: METHODOLOGY

3.1 System Architecture

The application follows a client-server architecture:

- Frontend: React.js, HTML, CSS, Tailwind CSS
- Backend: Node.js + Express.js
- Database: MongoDB
- AI Integration: OpenAI API

3.2 Data Handling

- User data stored securely in MongoDB
- Passwords hashed using bcrypt
- JWT used for authentication
- Travel plan data generated dynamically using AI

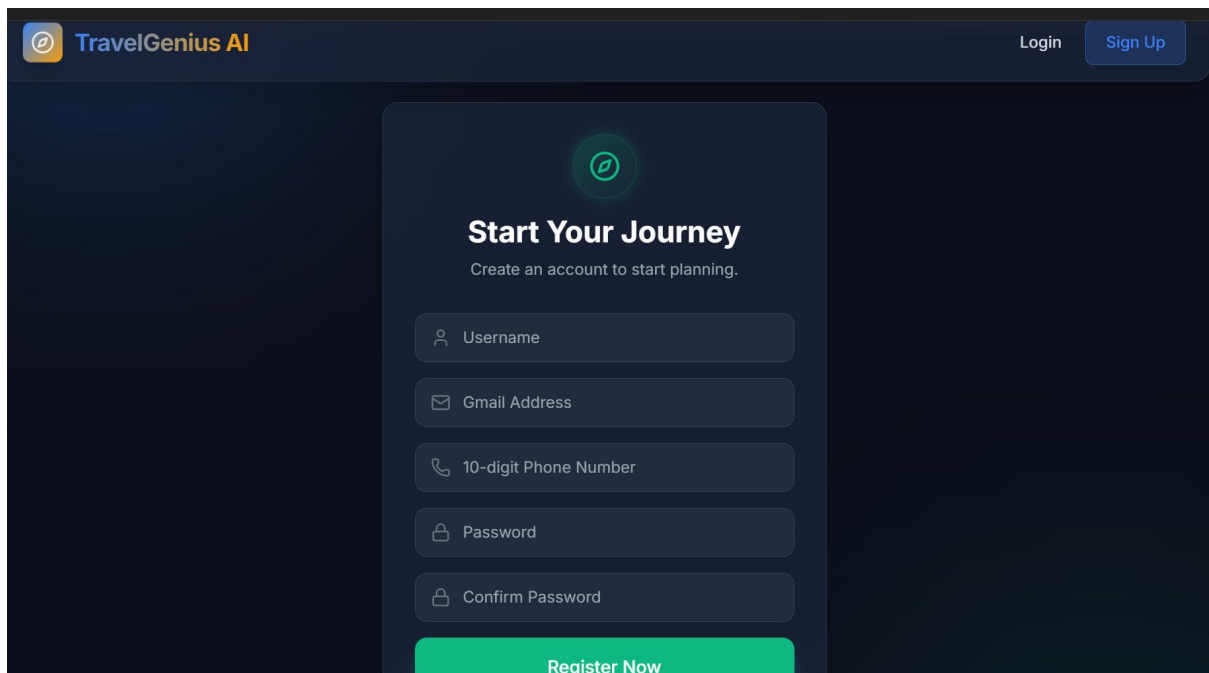
3.3 Functional Modules

- User Registration (Signup)
- User Login (Authentication)
- Plan Trip Module
- AI Travel Plan Generation
- Temples and History Module
- Famous Places Module
- Hotel Recommendation and Booking Module
- Restaurant Suggestion Module
- Email Notification Module

CHAPTER 4: RESULTS AND DISCUSSION

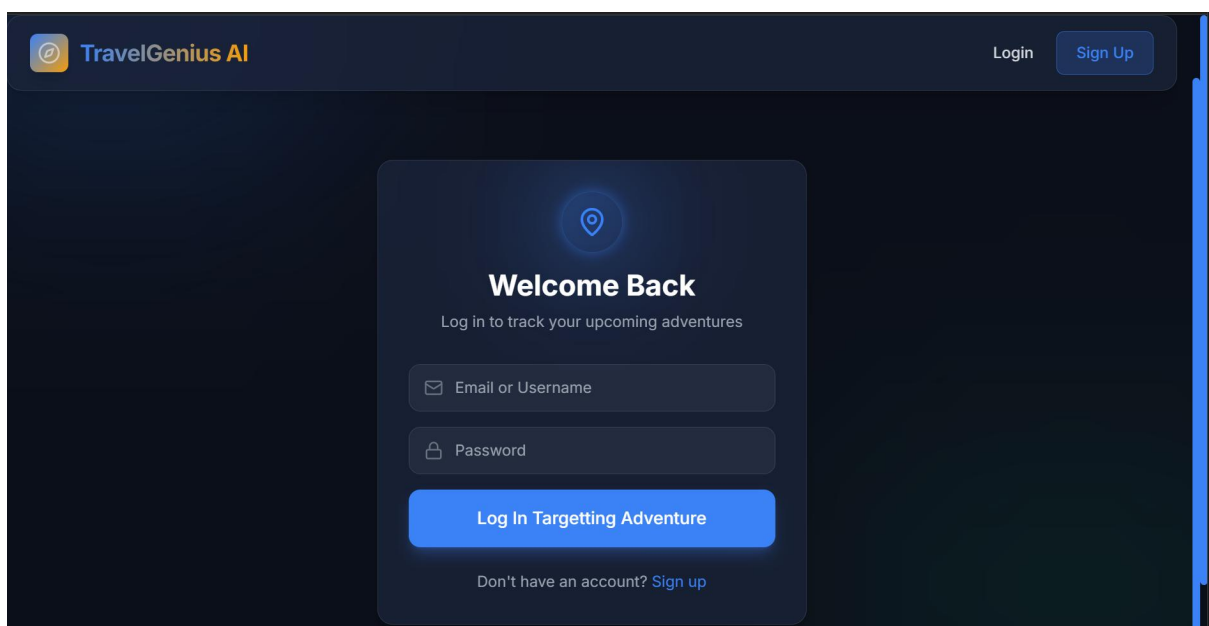
4.1 Implementation Results

- User Registration – SUCCESSFUL



The screenshot displays the user registration interface for TravelGenius AI. The header features the logo and navigation links for 'Login' and 'Sign Up'. The main content area is titled 'Start Your Journey' with the subtitle 'Create an account to start planning.' Below this, there are five input fields: 'Username', 'Email Address', '10-digit Phone Number', 'Password', and 'Confirm Password'. A green 'Register Now' button is positioned at the bottom of the form.

- User Login – SUCCESSFUL



The screenshot shows the user login interface for TravelGenius AI. The header includes the logo and navigation links for 'Login' and 'Sign Up'. The main content area is titled 'Welcome Back' with the subtitle 'Log in to track your upcoming adventures'. There are two input fields: 'Email or Username' and 'Password'. A blue 'Log In Targetting Adventure' button is located below the input fields. At the bottom, there is a link that says 'Don't have an account? Sign up'.

- Backend connected to MongoDB – SUCCESSFUL

```
58 console.error('❌ MONGO_URI is not defined in environment variables');
59 } else {
60 console.log('🐼 Attempting to connect to MongoDB... (URI length: ${mongoURI.length})');
61 }
```

Problems Output Debug Console **Terminal** Ports

PS C:\Users\Likitha\OneDrive\Desktop\AI TRAVEL PLANNER\backend> npm install

up to date, audited 183 packages in 1s

28 packages are looking for funding
run `npm fund` for details

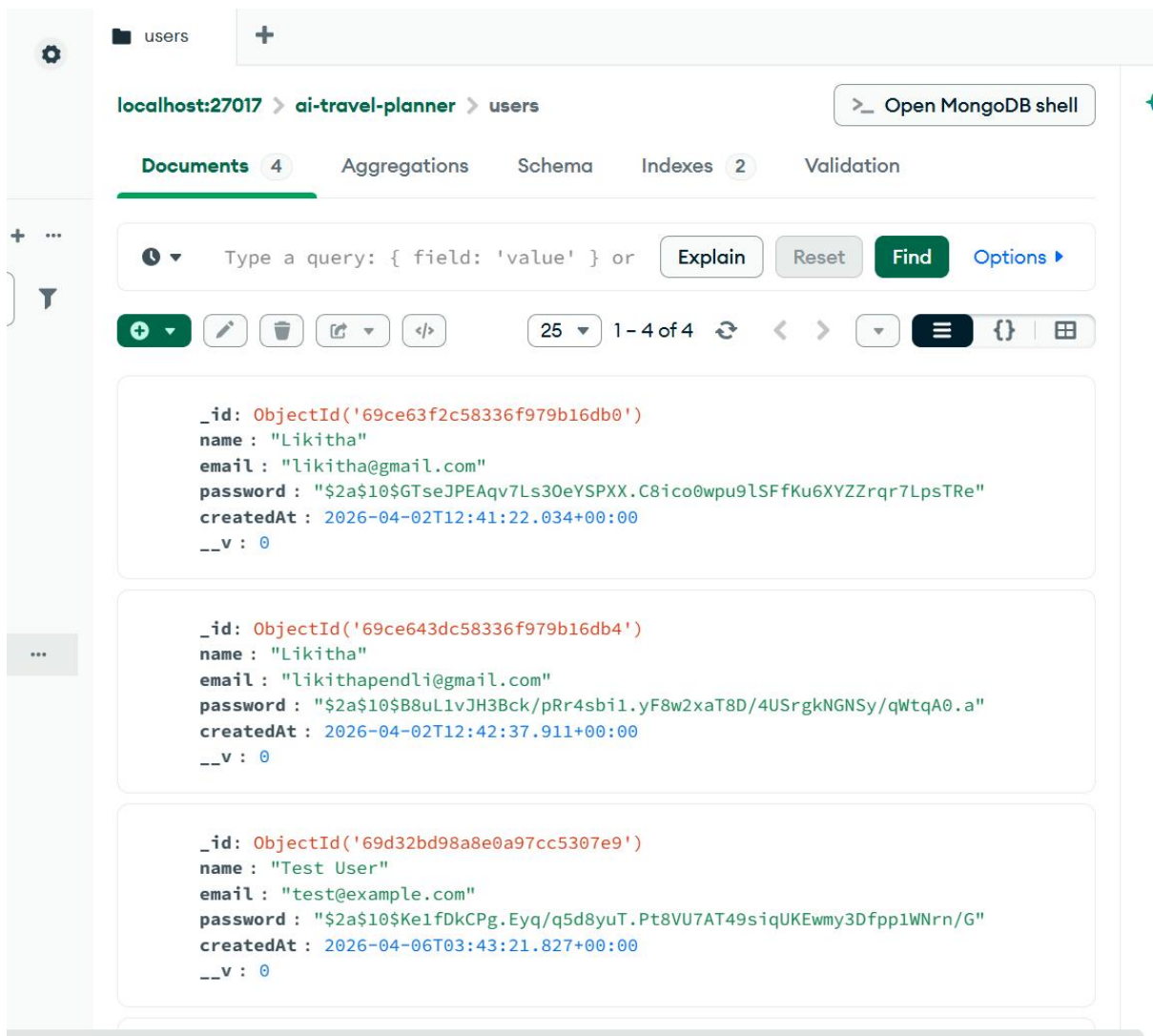
found 0 vulnerabilities

PS C:\Users\Likitha\OneDrive\Desktop\AI TRAVEL PLANNER\backend> node server.js

🐼 Attempting to connect to MongoDB... (URI length: 78)

✅ MongoDB connected

🚀 Server running on port 5000



- AI Travel Plan Generated – SUCCESSFUL

 **TravelGenius AI**

Plan Trip

Logout

Design Your Dream Trip

Our AI will craft the perfect itinerary optimized for your budget.

Where do you want to go in India?

e.g. Varanasi, Jaipur, Delhi...

Total Budget (₹ INR)

₹ e.g. 15000

Number of Days

e.g. 5

Generate Smart Plan

 **TravelGenius AI**

Plan Trip

Logout

Your Trip To Warangal

3 Days

₹3000 Total Budget

Send Trip Plan to Email

 **Budget Optimization**

Hotels (40%)

₹1200

Food (30%)

₹900

Travel & Misc (30%)

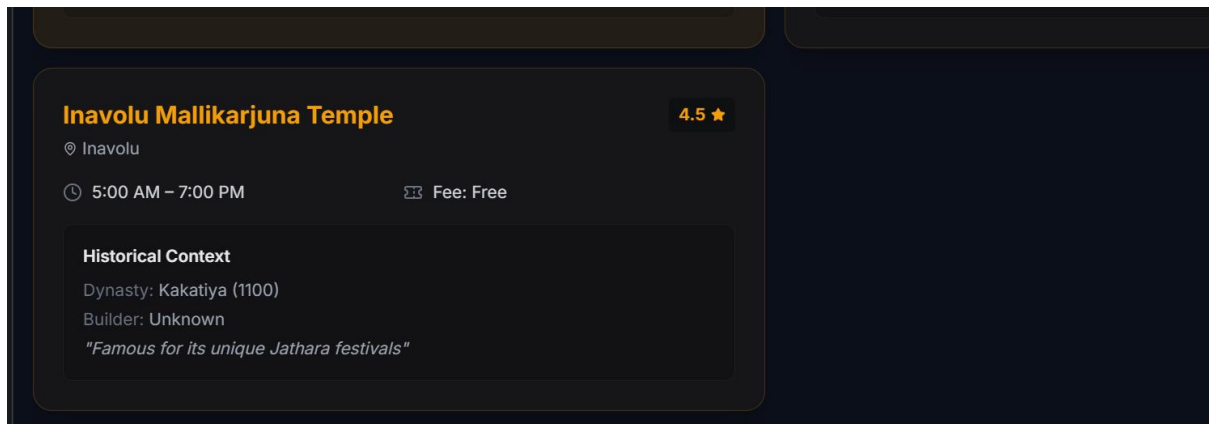
₹900

9

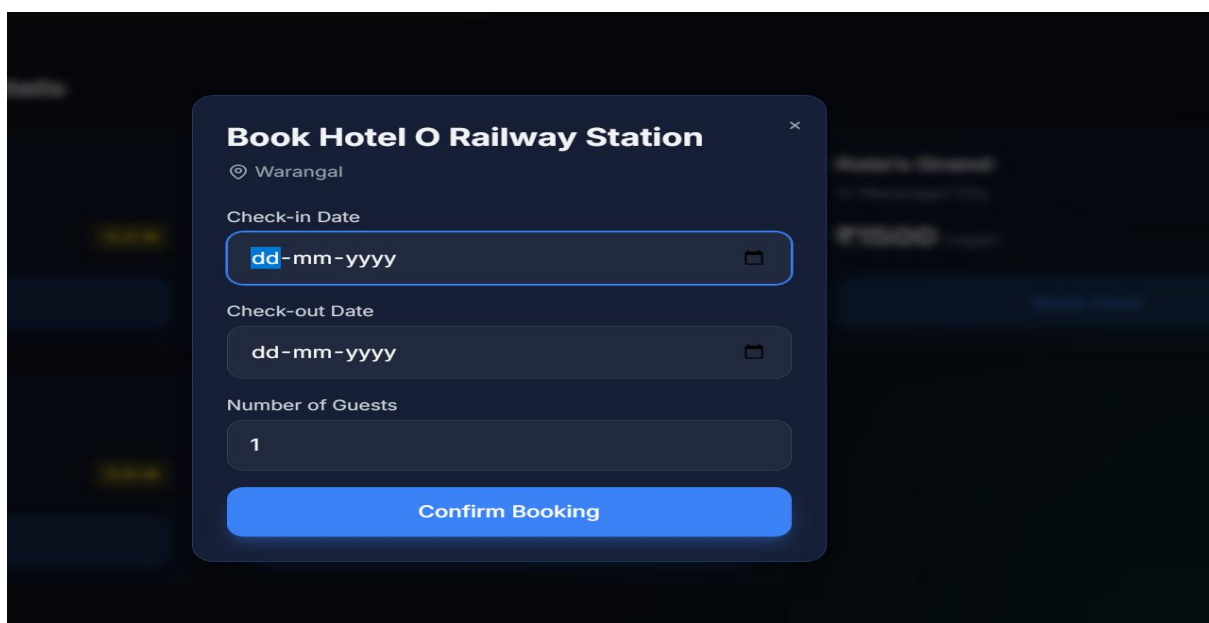
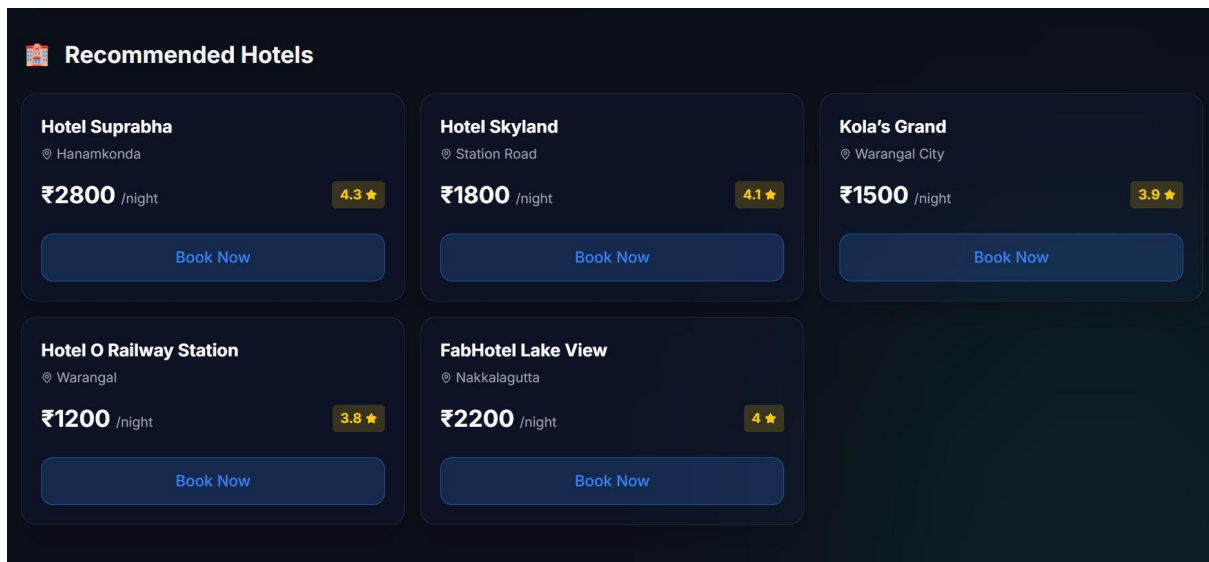
Daily Schedule		
Day 1	Day 2	Day 3
7 Locations Kush Mahal 📍 Warangal Fort LANDMARK Rayaparthi Shiva Temple 📍 Rayaparthi TEMPLE Jain Hill 📍 Padmakshi Gutta LANDMARK Bhadrakali Temple 📍 Warangal TEMPLE Pratap Nagar Market 📍 Warangal MARKET Bhadrakali Biodiversity Park 📍 Warangal PARK	7 Locations Wadepally Reservoir 📍 Hanamkonda LANDMARK Vana Vigyan Kendra Zoo 📍 Hanamkonda WILDLIFE Bheemuni Padam Waterfalls 📍 Gudur WATERFALL Station Road Shopping 📍 Warangal MARKET Ramappa Temple 📍 Palampet TEMPLE Warangal Arts College 📍 Subedari LANDMARK	7 Locations Nagnur Fort Ruins 📍 Nagnur LANDMARK Sri Vidya Saraswathi Temple 📍 Wargal TEMPLE Kazipet Junction (Historic) 📍 Kazipet LANDMARK Dharmasagar Reservoir 📍 Dharmasagar LANDMARK Mallanna Swamy Temple 📍 Komuravelli TEMPLE Sammakka Sarakka Jathara Path 📍 Medaram CULTURE

Top Famous Landmarks		
Warangal Fort 📍 Mathwada 13th-century fort featuring iconic stone arches (Kakatiya Toranas).	Laknavaram Lake 📍 Govindaraopet Scenic lake with a suspension bridge and island resorts.	Pakhal Lake & Wildlife Sanctuary 📍 Narsampet Stunning lake surrounded by lush green hilly terrain and wildlife.
Kakatiya Musical Garden 📍 Near Bhadrakali Temple A beautifully landscaped garden featuring a musical fountain.	Eturnagaram Wildlife Sanctuary 📍 Eturnagaram One of the oldest wildlife sanctuaries in Telangana.	

Spiritual & Historical Sites	
Thousand Pillar Temple 📍 Hanamkonda 🕒 6:00 AM – 8:00 PM 💰 Fee: Free Historical Context Dynasty: Kakatiya (1163) Builder: Rudra Deva <i>"Masterpiece of Kakatiya architecture with 1000 pillars"</i>	Ramappa Temple 📍 Palampet 🕒 6:00 AM – 6:00 PM 💰 Fee: Free Historical Context Dynasty: Kakatiya (1123) Builder: Recherla Rudra <i>"UNESCO World Heritage Site with floating bricks"</i>
Bhadrakali Temple 📍 Near Bhadrakali Lake 🕒 5:30 AM – 8:00 PM 💰 Fee: Free Historical Context Dynasty: Chalukya (625) Builder: Pulakeshin II <i>"One of the oldest temples for Goddess Bhadrakali"</i>	Padmakshi Temple 📍 Hanamkonda 🕒 6:00 AM – 6:00 PM 💰 Fee: Free Historical Context Dynasty: Kakatiya (1150) Builder: Prola II <i>"Jain and Hindu hybrid architecture on a hilltop"</i>



- Hotel Booking Feature – WORKING



🍴 Popular Restaurants

Green Bawarchi

Biryani

📍 Hanamkonda

4.2 ⭐

Khursheed Hotel

Hyderabadi

📍 Warangal City

4.5 ⭐

Green Park Biryani

Mughlai

📍 Subedari

4.1 ⭐

Skyv Pure Veg

South Indian

📍 Hanamkonda

4.3 ⭐

Mandi.com

Arabic

📍 Kazipet

4.4 ⭐



Contact Us

Have questions? Send us a message and we'll get back to you!

👤 Your Name

✉ Your Email

📞 Your Contact Number

➤ Submit

4.2 Key Features Achieved

- Full-stack integration
- AI-based travel planning
- Secure authentication system
- Budget optimization
- Temple history and cultural information
- Hotel booking system
- Email functionality

4.3 Discussion

The application successfully demonstrates how AI and full-stack development can be integrated to build an intelligent travel planning system. It reduces manual effort and provides users with a structured and efficient way to plan trips.

Future enhancements can include real-time booking APIs, map integration, and advanced recommendation systems.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

The “AI Travel Planner with Budget Optimizer” project successfully demonstrates the development of a smart travel planning web application using full-stack technologies and artificial intelligence. The system integrates a responsive frontend with a robust backend, ensuring secure authentication and efficient data management.

The application provides users with personalized travel plans, including temples with history, famous places, hotels, and restaurants. It also includes budget optimization and booking features, making it a complete travel solution.

Overall, the project reflects modern web development practices and highlights the importance of AI in enhancing user experience.

5.2 Future Scope

The system can be further enhanced with:

- Integration of Google Maps API for navigation and route guidance
- Real-time hotel booking and pricing APIs
- AI chatbot for travel assistance
- Mobile application development
- Deployment on cloud platforms (AWS, Vercel, Render)
- Multi-language support for wider accessibility